

Assignment 10: Data Scraping

Shiyi Zheng

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1
library(tidyverse)
library(rvest)
library(here)
library(lubridate)
here()
```

```
## [1] "/home/guest/EDE_Spring2026_Shiiyi"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):
 - Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
 - Set the year to 2024
 - Scroll down and select the LWSP link next to Durham Municipality.
 - Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwdid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

#2

```
url_durham_2024 <- "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024"
web_durham_2024 <- read_html(url_durham_2024)
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
 - Water system name
 - Contact Person
 - Ownership
- From the “3. Water Supply Sources” section:
 - Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “Reginald Hicks”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

#3

```
Water_system_name <- web_durham_2024 %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
  html_text()
Water_system_name
```

```
## [1] "Durham"
```

```
Contact_person <- web_durham_2024 %>%
  html_nodes("div+ table tr:nth-child(4) td:nth-child(2)") %>%
  html_text()
Contact_person
```

```
## [1] "Reginald Hicks"
```

```
Ownership <- web_durham_2024 %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
  html_text()
Ownership
```

```
## [1] "Municipality"
```

```
MGD_each_month <- web_durham_2024 %>%
  html_nodes("th~ td+ td") %>%
  html_text() %>%
  as.numeric()
MGD_each_month
```

```
## [1] 34.50 36.06 37.33 32.10 46.65 37.36 38.20 41.90 36.58 36.73 42.96 34.45
```

```
length(MGD_each_month)
```

```
## [1] 12
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It's likely you won't be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```
#4

df_durham_2024 <- data.frame("Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
                             "Year" = rep(2024, 12),
                             "Maximum_Day_Use_MGD" = as.numeric(MGD_each_month))

df_durham_2024 <- df_durham_2024 %>%
  mutate(Water_System_Name=!!Water_system_name,
         Contact_Person=!!Contact_person,
         Ownership=!!Ownership,
         Date=my(paste(Month,"-",Year))
        ) %>%
  arrange(Date)
df_durham_2024
```

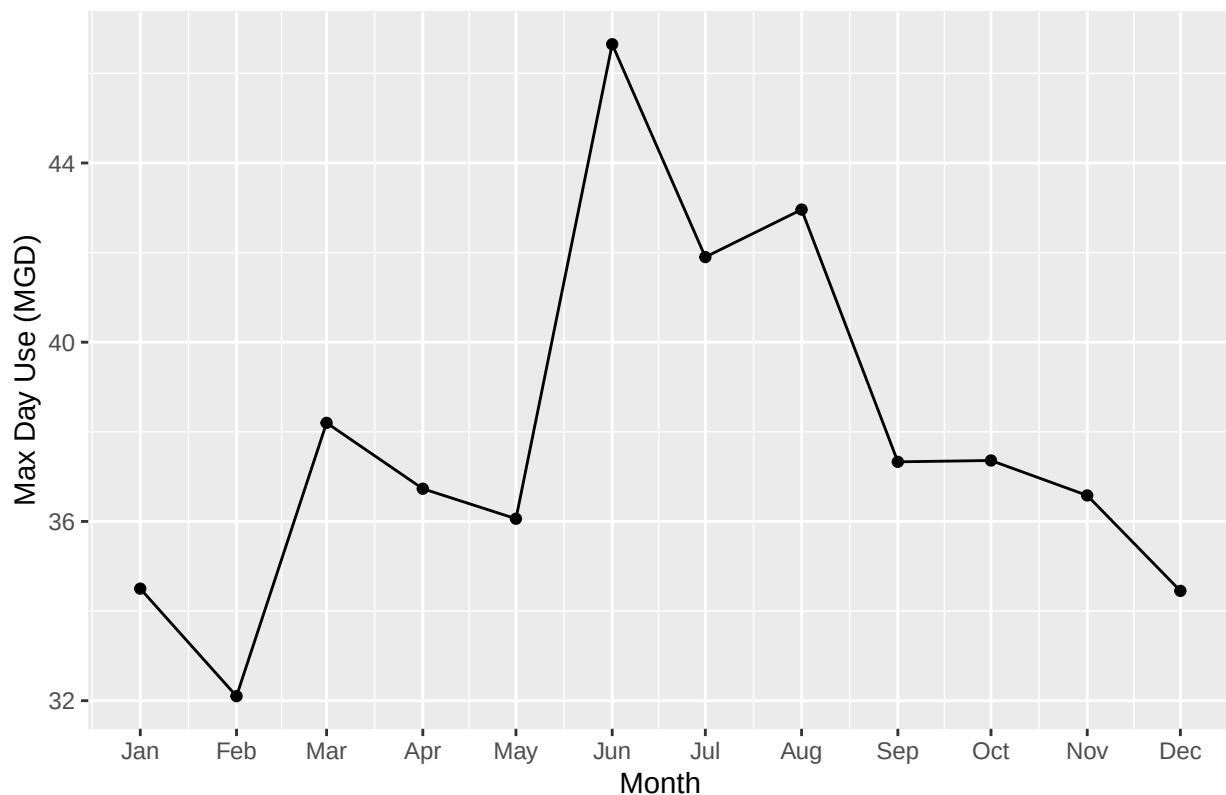
```
##      Month Year Maximum_Day_Use_MGD Water_System_Name Contact_Person Ownership
## 1         1 2024             34.50      Durham Reginald Hicks Municipality
## 2         2 2024             32.10      Durham Reginald Hicks Municipality
## 3         3 2024             38.20      Durham Reginald Hicks Municipality
## 4         4 2024             36.73      Durham Reginald Hicks Municipality
## 5         5 2024             36.06      Durham Reginald Hicks Municipality
## 6         6 2024             46.65      Durham Reginald Hicks Municipality
## 7         7 2024             41.90      Durham Reginald Hicks Municipality
## 8         8 2024             42.96      Durham Reginald Hicks Municipality
## 9         9 2024             37.33      Durham Reginald Hicks Municipality
## 10        10 2024             37.36      Durham Reginald Hicks Municipality
## 11        11 2024             36.58      Durham Reginald Hicks Municipality
## 12        12 2024             34.45      Durham Reginald Hicks Municipality
##           Date
## 1 2024-01-01
## 2 2024-02-01
## 3 2024-03-01
## 4 2024-04-01
```

```
## 5 2024-05-01
## 6 2024-06-01
## 7 2024-07-01
## 8 2024-08-01
## 9 2024-09-01
## 10 2024-10-01
## 11 2024-11-01
## 12 2024-12-01
```

```
#5
```

```
ggplot(df_durham_2024, aes(x=Date, y=Maximum_Day_Use_MGD)) +
  geom_line() +
  geom_point() +
  scale_x_date(date_breaks = "1 month", date_labels = "%b") +
  labs(title = "Water Supply Sources of Max Day Use (MGD) in NC Durham (2024) ",
       x="Month",
       y="Max Day Use (MGD)")
```

Water Supply Sources of Max Day Use (MGD) in NC Durham (2024)



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs, “PWSID” and “year”, that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)

- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function's output

```
#6.
Scrape_LWSP <- function(the_PWSID, the_year){
  the_base_url <- "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid="
  the_scrape_url <- paste0(the_base_url, the_PWSID, "&year=", the_year)
  print(the_scrape_url)

  web_LWSP <- read_html(the_scrape_url)
  print(web_LWSP)

  Water_system_name_tag <- "div+ table tr:nth-child(1) td:nth-child(2)"
  Contact_person_tag <- "div+ table tr:nth-child(4) td:nth-child(2)"
  Ownership_tag <- "div+ table tr:nth-child(2) td:nth-child(4)"
  MGD_each_month_tag <- "th~ td+ td"

  Water_system_name <- web_LWSP%>%
    html_nodes(Water_system_name_tag)%>%
    html_text()%>%
    str_trim()
  Contact_person <- web_LWSP %>%
    html_nodes(Contact_person_tag)%>%
    html_text()%>%
    str_trim()
  Ownership <- web_LWSP %>%
    html_nodes(Ownership_tag)%>%
    html_text()%>%
    str_trim()
  MGD_each_month <- web_LWSP %>%
    html_nodes(MGD_each_month_tag)%>%
    html_text()%>%
    str_trim()%>%
    as.numeric()

  df_LWSP <-data.frame("Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
                      "Year" = rep(the_year, 12),
                      "PWSID" = rep(the_PWSID, 12),
                      "Maximum_Day_Use_MGD" = as.numeric(MGD_each_month)) %>%
    mutate(Water_System_Name = !!Water_system_name,
           Contact_Person = !!Contact_person,
           Ownership = !!Ownership,
           Date = my(paste(Month, "-", Year)))%>%
    arrange(Date)

  return(df_LWSP)
}
```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```
#7
df_Durham_2020 <- Scrape_LWSP('03-32-010',2020)

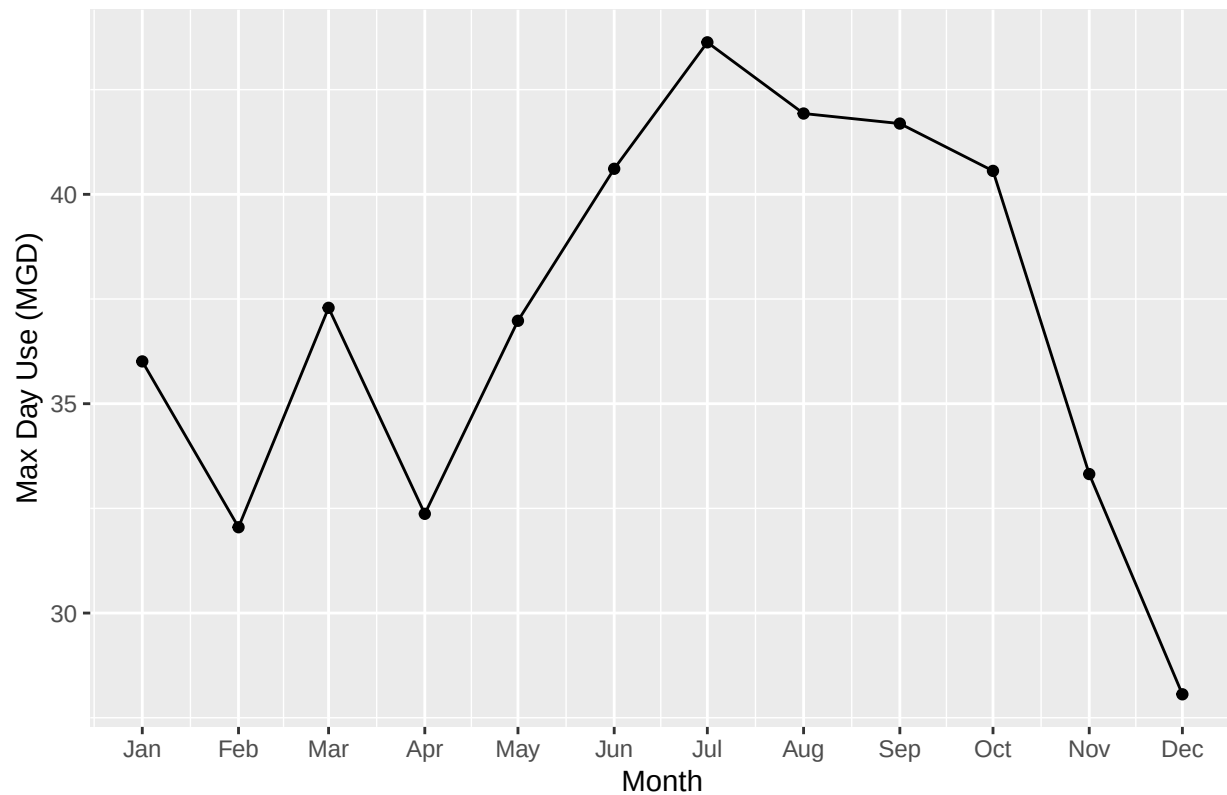
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-32-010&year=2020"
## {html_document}
## <html xmlns="http://www.w3.org/1999/xhtml" lang="en" xml:lang="en">
## [1] <head>\n<title>DWR :: Local Water Supply Planning</title>\n<meta http-equiv= ...
## [2] <body id="plan">\r\n<!--<div id="division-header">\r\n<a name="top" href= ...

print(df_Durham_2020)
```

```
##      Month Year      PWSID Maximum_Day_Use_MGD Water_System_Name Contact_Person
## 1      1 2020 03-32-010          36.01          Durham Reginald Hicks
## 2      2 2020 03-32-010          32.05          Durham Reginald Hicks
## 3      3 2020 03-32-010          37.29          Durham Reginald Hicks
## 4      4 2020 03-32-010          32.37          Durham Reginald Hicks
## 5      5 2020 03-32-010          36.98          Durham Reginald Hicks
## 6      6 2020 03-32-010          40.61          Durham Reginald Hicks
## 7      7 2020 03-32-010          43.63          Durham Reginald Hicks
## 8      8 2020 03-32-010          41.93          Durham Reginald Hicks
## 9      9 2020 03-32-010          41.69          Durham Reginald Hicks
## 10     10 2020 03-32-010          40.56          Durham Reginald Hicks
## 11     11 2020 03-32-010          33.32          Durham Reginald Hicks
## 12     12 2020 03-32-010          28.06          Durham Reginald Hicks
##      Ownership      Date
## 1 Municipality 2020-01-01
## 2 Municipality 2020-02-01
## 3 Municipality 2020-03-01
## 4 Municipality 2020-04-01
## 5 Municipality 2020-05-01
## 6 Municipality 2020-06-01
## 7 Municipality 2020-07-01
## 8 Municipality 2020-08-01
## 9 Municipality 2020-09-01
## 10 Municipality 2020-10-01
## 11 Municipality 2020-11-01
## 12 Municipality 2020-12-01
```

```
ggplot(df_Durham_2020,aes(x=Date,y=Maximum_Day_Use_MGD))+
  geom_line()+
  geom_point()+
  scale_x_date(date_breaks = "1 month", date_labels = "%b") +
  labs(title = "Water Supply Sources of Max Day Use (MGD) in NC Durham (2020) ",
       x="Month",
       y="Max Day Use (MGD)")
```

Water Supply Sources of Max Day Use (MGD) in NC Durham (2020)



```
str(df_Durham_2020)
```

```
## 'data.frame':    12 obs. of  8 variables:
##  $ Month          : num  1 2 3 4 5 6 7 8 9 10 ...
##  $ Year            : num  2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
##  $ PWSID           : chr  "03-32-010" "03-32-010" "03-32-010" "03-32-010" ...
##  $ Maximum_Day_Use_MGD: num  36 32 37.3 32.4 37 ...
##  $ Water_System_Name : chr  "Durham" "Durham" "Durham" "Durham" ...
##  $ Contact_Person   : chr  "Reginald Hicks" "Reginald Hicks" "Reginald Hicks" "Reginald Hicks" ...
##  $ Ownership        : chr  "Municipality" "Municipality" "Municipality" "Municipality" ...
##  $ Date             : Date, format: "2020-01-01" "2020-02-01" ...
```

```
glimpse(df_Durham_2020)
```

```
## Rows: 12
## Columns: 8
##  $ Month          <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
##  $ Year            <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 20~
##  $ PWSID           <chr> "03-32-010", "03-32-010", "03-32-010", "03-32-010"~
##  $ Maximum_Day_Use_MGD <dbl> 36.01, 32.05, 37.29, 32.37, 36.98, 40.61, 43.63, 4~
##  $ Water_System_Name <chr> "Durham", "Durham", "Durham", "Durham", "Durham", ~
##  $ Contact_Person   <chr> "Reginald Hicks", "Reginald Hicks", "Reginald Hick~
##  $ Ownership        <chr> "Municipality", "Municipality", "Municipality", "M~
##  $ Date             <date> 2020-01-01, 2020-02-01, 2020-03-01, 2020-04-01, 20~
```

8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```
#8
df_Cary_2020 <- Scrape_LWSP('03-92-020',2020)

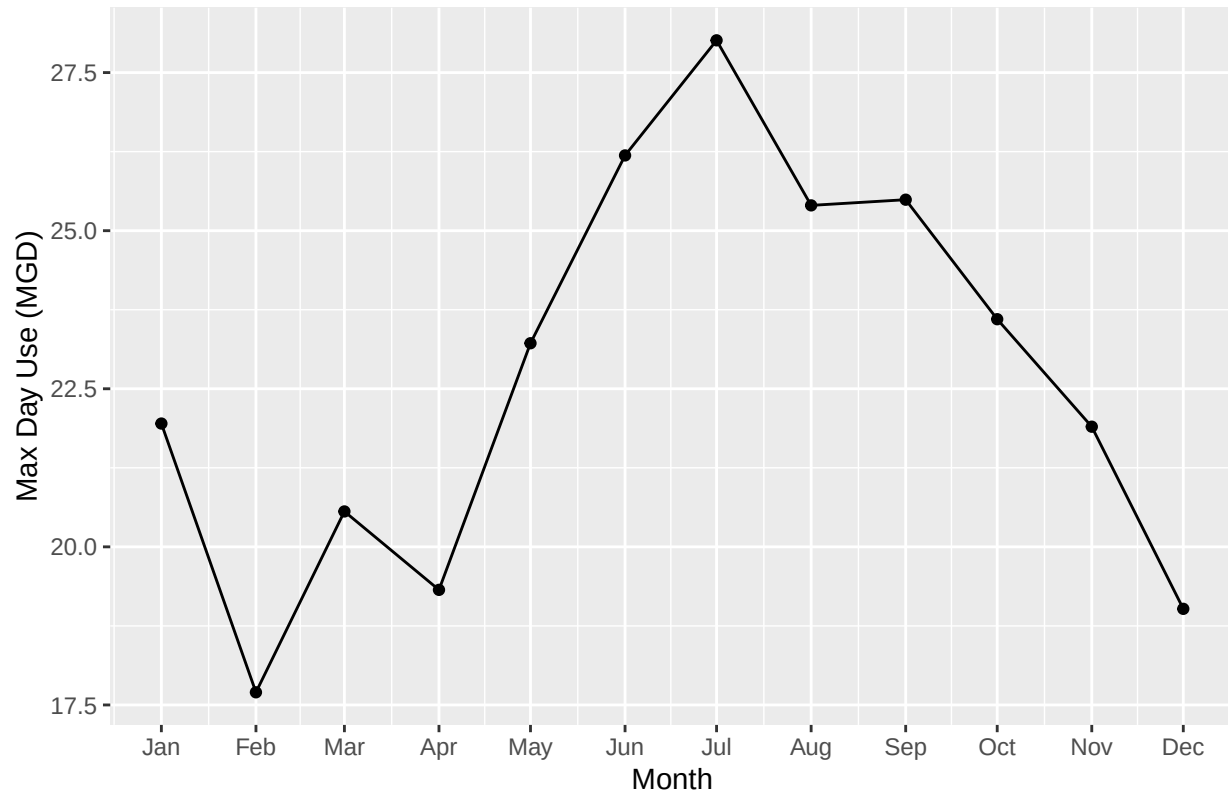
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-92-020&year=2020"
## {html_document}
## <html xmlns="http://www.w3.org/1999/xhtml" lang="en" xml:lang="en">
## [1] <head>\n<title>DWR :: Local Water Supply Planning</title>\n<meta http-equ ...
## [2] <body id="plan">\r\n<!--<div id="division-header">\r\n<a name="top" href= ...
```

```
print(df_Cary_2020)
```

```
##      Month Year      PWSID Maximum_Day_Use_MGD Water_System_Name Contact_Person
## 1      1 2020 03-92-020          21.95          Cary      Jamie Revels
## 2      2 2020 03-92-020          17.70          Cary      Jamie Revels
## 3      3 2020 03-92-020          20.56          Cary      Jamie Revels
## 4      4 2020 03-92-020          19.32          Cary      Jamie Revels
## 5      5 2020 03-92-020          23.22          Cary      Jamie Revels
## 6      6 2020 03-92-020          26.19          Cary      Jamie Revels
## 7      7 2020 03-92-020          28.01          Cary      Jamie Revels
## 8      8 2020 03-92-020          25.40          Cary      Jamie Revels
## 9      9 2020 03-92-020          25.49          Cary      Jamie Revels
## 10     10 2020 03-92-020          23.60          Cary      Jamie Revels
## 11     11 2020 03-92-020          21.90          Cary      Jamie Revels
## 12     12 2020 03-92-020          19.02          Cary      Jamie Revels
##      Ownership      Date
## 1 Municipality 2020-01-01
## 2 Municipality 2020-02-01
## 3 Municipality 2020-03-01
## 4 Municipality 2020-04-01
## 5 Municipality 2020-05-01
## 6 Municipality 2020-06-01
## 7 Municipality 2020-07-01
## 8 Municipality 2020-08-01
## 9 Municipality 2020-09-01
## 10 Municipality 2020-10-01
## 11 Municipality 2020-11-01
## 12 Municipality 2020-12-01
```

```
ggplot(df_Cary_2020,aes(x=Date,y=Maximum_Day_Use_MGD))+
  geom_line()+
  geom_point()+
  scale_x_date(date_breaks = "1 month", date_labels = "%b") +
  labs(title = "Water Supply Sources of Max Day Use (MGD) in NC Cary (2020) ",
       x="Month",
       y="Max Day Use (MGD)")
```


Water Supply Sources of Max Day Use (MGD) in NC Cary (2020)



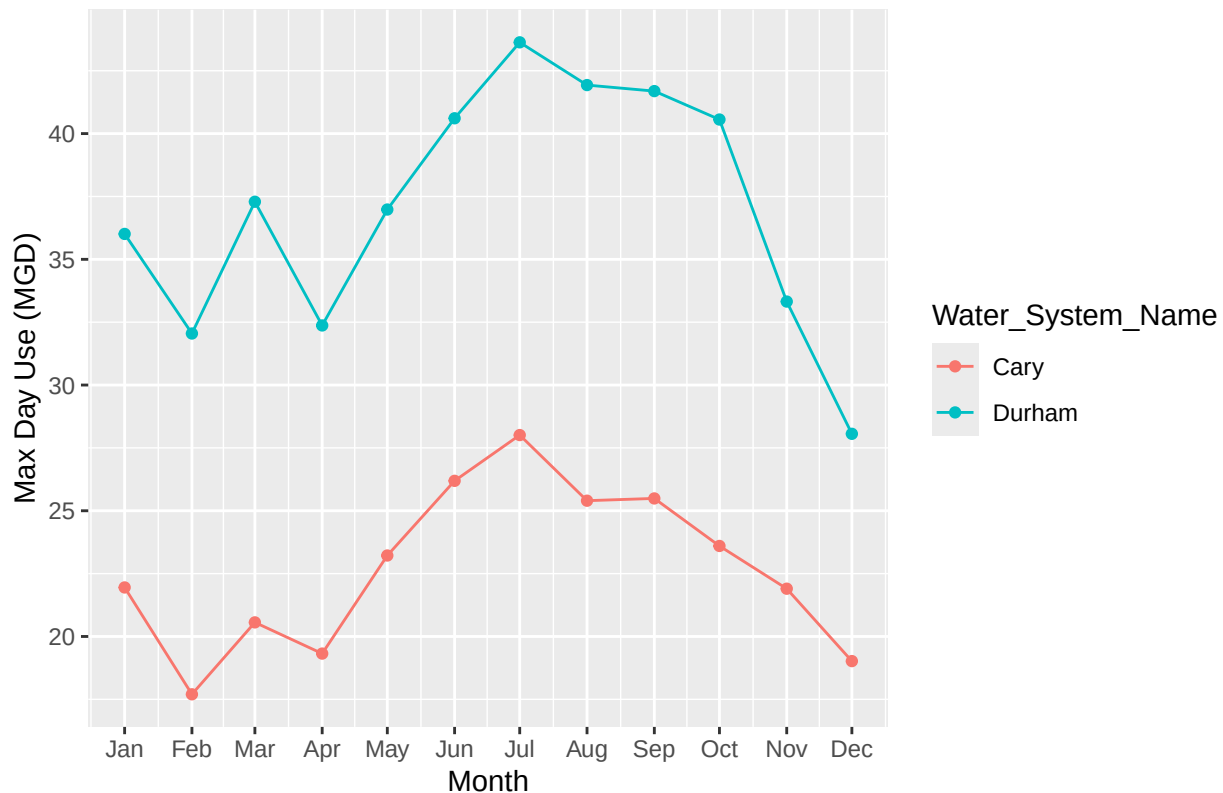
```
df_Durham_Cary_2020 <- bind_rows(df_Durham_2020,df_Cary_2020)
df_Durham_Cary_2020
```

##	Month	Year	PWSID	Maximum_Day_Use_MGD	Water_System_Name	Contact_Person
## 1	1	2020	03-32-010	36.01	Durham	Reginald Hicks
## 2	2	2020	03-32-010	32.05	Durham	Reginald Hicks
## 3	3	2020	03-32-010	37.29	Durham	Reginald Hicks
## 4	4	2020	03-32-010	32.37	Durham	Reginald Hicks
## 5	5	2020	03-32-010	36.98	Durham	Reginald Hicks
## 6	6	2020	03-32-010	40.61	Durham	Reginald Hicks
## 7	7	2020	03-32-010	43.63	Durham	Reginald Hicks
## 8	8	2020	03-32-010	41.93	Durham	Reginald Hicks
## 9	9	2020	03-32-010	41.69	Durham	Reginald Hicks
## 10	10	2020	03-32-010	40.56	Durham	Reginald Hicks
## 11	11	2020	03-32-010	33.32	Durham	Reginald Hicks
## 12	12	2020	03-32-010	28.06	Durham	Reginald Hicks
## 13	1	2020	03-92-020	21.95	Cary	Jamie Revels
## 14	2	2020	03-92-020	17.70	Cary	Jamie Revels
## 15	3	2020	03-92-020	20.56	Cary	Jamie Revels
## 16	4	2020	03-92-020	19.32	Cary	Jamie Revels
## 17	5	2020	03-92-020	23.22	Cary	Jamie Revels
## 18	6	2020	03-92-020	26.19	Cary	Jamie Revels
## 19	7	2020	03-92-020	28.01	Cary	Jamie Revels
## 20	8	2020	03-92-020	25.40	Cary	Jamie Revels
## 21	9	2020	03-92-020	25.49	Cary	Jamie Revels

## 22	10	2020	03-92-020	23.60	Cary	Jamie Revels
## 23	11	2020	03-92-020	21.90	Cary	Jamie Revels
## 24	12	2020	03-92-020	19.02	Cary	Jamie Revels
##			Ownership	Date		
## 1			Municipality	2020-01-01		
## 2			Municipality	2020-02-01		
## 3			Municipality	2020-03-01		
## 4			Municipality	2020-04-01		
## 5			Municipality	2020-05-01		
## 6			Municipality	2020-06-01		
## 7			Municipality	2020-07-01		
## 8			Municipality	2020-08-01		
## 9			Municipality	2020-09-01		
## 10			Municipality	2020-10-01		
## 11			Municipality	2020-11-01		
## 12			Municipality	2020-12-01		
## 13			Municipality	2020-01-01		
## 14			Municipality	2020-02-01		
## 15			Municipality	2020-03-01		
## 16			Municipality	2020-04-01		
## 17			Municipality	2020-05-01		
## 18			Municipality	2020-06-01		
## 19			Municipality	2020-07-01		
## 20			Municipality	2020-08-01		
## 21			Municipality	2020-09-01		
## 22			Municipality	2020-10-01		
## 23			Municipality	2020-11-01		
## 24			Municipality	2020-12-01		

```
ggplot(df_Durham_Cary_2020,aes(x=Date,y=Maximum_Day_Use_MGD,color=Water_System_Name))+
  geom_line()+
  geom_point()+
  scale_x_date(date_breaks = "1 month", date_labels = "%b") +
  labs(title = "Water Supply Sources of Max Day Use (MGD): Durham vs Cary in NC (2020) ",
        x="Month",
        y="Max Day Use (MGD)")
```

Water Supply Sources of Max Day Use (MGD): Durham vs Cary in NC (202



- Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

TIP: See Section 3.2 in the "06_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs (where one of the two inputs does not change). Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

#9

```
years_range <- c(2018:2023)
PWSID_Cary <- rep('03-92-020', length(years_range))
library(purrr)
df_Cary_2018_2023 <- map2(PWSID_Cary, years_range, Scrape_LWSP)

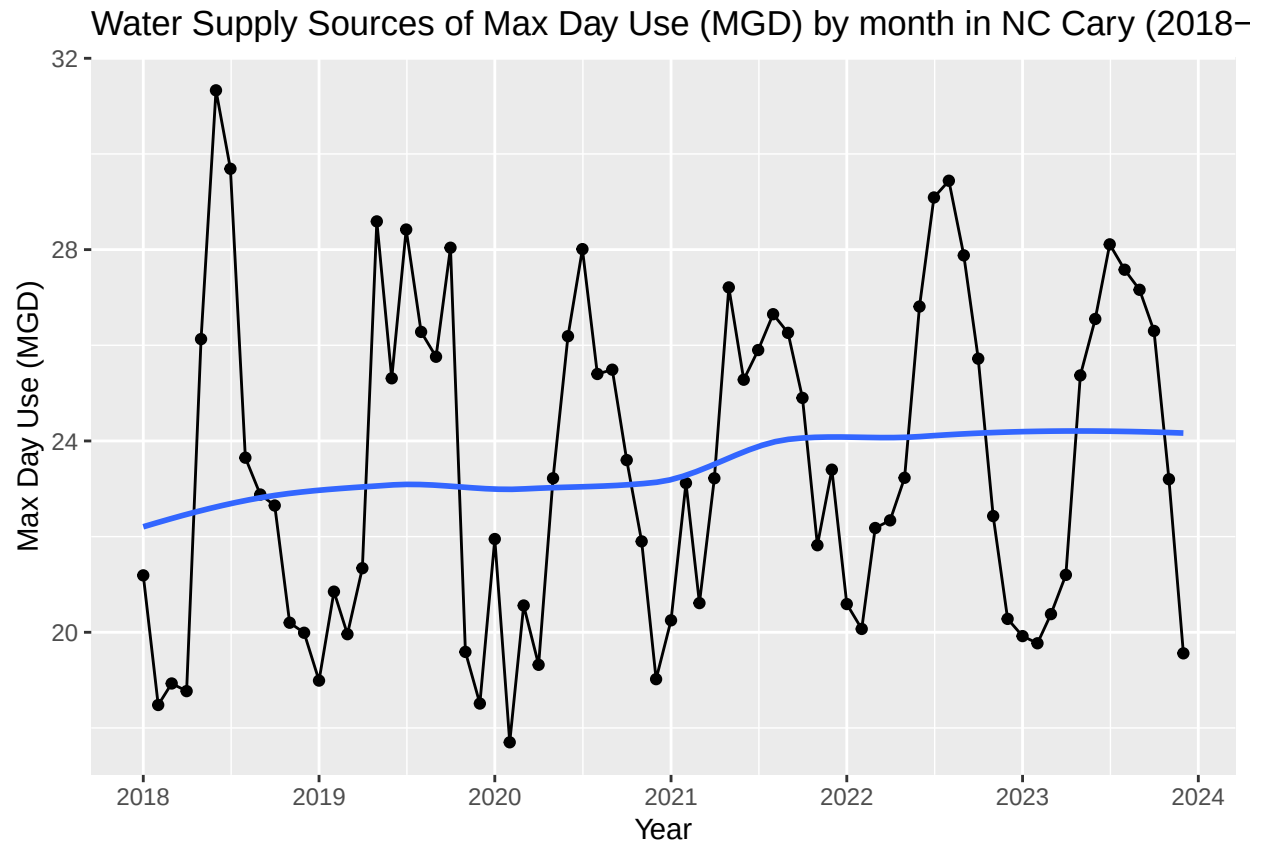
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-92-020&year=2018"
## {html_document}
## <html xmlns="http://www.w3.org/1999/xhtml" lang="en" xml:lang="en">
## [1] <head>\n<title>DWR :: Local Water Supply Planning</title>\n<meta http-equ ...
## [2] <body id="plan">\r\n<!--<div id="division-header">\r\n<a name="top" href= ...
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-92-020&year=2019"
## {html_document}
## <html xmlns="http://www.w3.org/1999/xhtml" lang="en" xml:lang="en">
## [1] <head>\n<title>DWR :: Local Water Supply Planning</title>\n<meta http-equ ...
## [2] <body id="plan">\r\n<!--<div id="division-header">\r\n<a name="top" href= ...
```

```
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-92-020&year=2020"
## {html_document}
## <html xmlns="http://www.w3.org/1999/xhtml" lang="en" xml:lang="en">
## [1] <head>\n<title>DWR :: Local Water Supply Planning</title>\n<meta http-equ ...
## [2] <body id="plan">\r\n<!--<div id="division-header">\r\n<a name="top" href= ...
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-92-020&year=2021"
## {html_document}
## <html xmlns="http://www.w3.org/1999/xhtml" lang="en" xml:lang="en">
## [1] <head>\n<title>DWR :: Local Water Supply Planning</title>\n<meta http-equ ...
## [2] <body id="plan">\r\n<!--<div id="division-header">\r\n<a name="top" href= ...
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-92-020&year=2022"
## {html_document}
## <html xmlns="http://www.w3.org/1999/xhtml" lang="en" xml:lang="en">
## [1] <head>\n<title>DWR :: Local Water Supply Planning</title>\n<meta http-equ ...
## [2] <body id="plan">\r\n<!--<div id="division-header">\r\n<a name="top" href= ...
## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-92-020&year=2023"
## {html_document}
## <html xmlns="http://www.w3.org/1999/xhtml" lang="en" xml:lang="en">
## [1] <head>\n<title>DWR :: Local Water Supply Planning</title>\n<meta http-equ ...
## [2] <body id="plan">\r\n<!--<div id="division-header">\r\n<a name="top" href= ...
```

```
df_Cary_2018_2023_dataframe <- bind_rows(df_Cary_2018_2023)

ggplot(df_Cary_2018_2023_dataframe,aes(x=Date,y=Maximum_Day_Use_MGD))+
  geom_line()+
  geom_point()+
  geom_smooth(method = "loess", se = FALSE)+
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
  labs(title = "Water Supply Sources of Max Day Use (MGD) by month in NC Cary (2018-2023) ",
       x="Year",
       y="Max Day Use (MGD)")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: Yes. The smoothed trend line suggests a modest but consistent increase in maximum daily water withdrawals over the 2018–2023 period. While seasonal fluctuations are evident, the long-run pattern indicates gradual growth in water demand. >